MagicCode Super-Resolution Software Known Issues v1.1.0

Document Version: v1.1.0

1. Scope

This document lists known limitations, constraints, and integration risks for MagicCode Super-Resolution Software v1.1.0. It should be reviewed before release, integration, and customer deployment.

2. Functional Constraints

- Input width and height must be within 64 to 4032 pixels.
- The requested scaling factor must be within 1.0 to 8.0, but actual availability depends on backend and model configuration.
- CPU backends may support only implemented integer scale paths for some configurations.
- Model files must match backend, mode, and scale expectations. A mismatched model can cause initialization failure.
- Super-resolution output quality depends heavily on input quality, noise level, compression, motion, and content type.

3. Backend Constraints

- CPU backends use buffer input and require supported CPU features such as AVX2 or NEON.
- GPU backends use texture input and require a valid graphics context/device and compatible driver.
- Vulkan and OpenGLES behavior may vary across Android devices and GPU vendors.
- Metal behavior may vary across iOS/iPadOS/macOS devices and OS versions.

4. Engine Integration Constraints

- Unity Android requires the native plugin under Assets/Plugins/Android/arm64-v8a/.
- Unity iOS requires static library linkage in the generated Xcode target.
- Unreal Engine Android packaging may require a compatible NDK version for older UE versions.
- Unreal Engine iOS packaging depends on matching UE, Xcode, and Apple SDK versions.

5. Operational Constraints

- Performance varies by device hardware, backend, input resolution, scale, mode, thread count, and thermal state.
- Multi-threading is useful for larger inputs but may add overhead for smaller inputs.
- Repeated initialization and uninitialization can be more expensive than reusing a handle or using MC_Control where appropriate.
- Network restrictions may prevent technical reporting from reaching MagicCode servers; processing should not rely on reporting success.

6. Recommended Mitigation

- Validate each target platform and device class with production-like content.
- Use MC_Control(QUERY_STATUS) during integration tests.
- Log error codes and map them to the Error Code Manual.
- Package only the model files required by the selected backend and mode.
- Maintain a release-specific compatibility matrix for customers.